Feasibility study of determining charm mixing and CP violation parameters in super tau-charm factory

AAAA [1,2], BBB [1,2], CCC [1,2]
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Based on the QCFilterAlg package, a new Gaudi-based package, for quantum-coherent correction of $D^0\bar{D}^0$ simulation samples, is developed under the BOSS software framework to improve the $D^0\bar{D}^0$ simulation at BESIII.

I. INTRODUCTION

In the neutral charm meson system, the initially produced flavour eigenstate $D^0/\bar{D}^0$ oscillates to $\bar{D}^0/D^0$ before it decays as is known as the charm mixing. The physical mass eigenstates can be expressed as superpositions of the flavour eigenstates

$$\begin{aligned} |D_1\rangle &= p\,|D^0\rangle + q\,|\bar{D}^0\rangle\,, \\ |D_2\rangle &= p\,|D^0\rangle - q\,|\bar{D}^0\rangle\,. \end{aligned} \tag{1}$$

Here $p(q)$ are complex numbers that quantify the size of CP violation effects and can be normalized with $|p|^2+|q|^2=1$. In the convention $CP\,|D^0\rangle = |\bar{D}^0\rangle$ and $CP\,|\bar{D}^0\rangle = |D^0\rangle$, $|D_{1(2)}\rangle$ are CP-even(odd) eigenstates assuming there was no CP violation. Time evolution of the initial flavour eigenstate is given by:

$$\begin{aligned} |D^0(t)\rangle &= g_+(t)\,|D^0\rangle + \tfrac{q}{p} g_-(t)\,|\bar{D}^0\rangle\,, \\ |\bar{D}^0(t)\rangle &= g_+(t)\,|\bar{D}^0\rangle + \tfrac{p}{q} g_-(t)\,|D^0\rangle\,. \end{aligned} \tag{2}$$

It is derived from time evolution of the mass eigenstates $|D_{1(2)}\rangle$, which is

$$\begin{aligned} |D_1(t)\rangle &= \exp\left(-i\mu_1 t\right)|D_1\rangle = \exp\left(-i(m_1 - \tfrac{i}{2}\Gamma_1)t\right)|D_1\rangle\,, \\ |D_2(t)\rangle &= \exp\left(-i\mu_2 t\right)|D_2\rangle = \exp\left(-i(m_2 - \tfrac{i}{2}\Gamma_2)t\right)|D_2\rangle\,. \end{aligned} \tag{3}$$

Therefore, with Eq. 1 the coefficients $g_\pm(t)$ take the form of

$$g_\pm(t) = \frac{1}{2}\left[\exp\left(-i(m_1 - \frac{i}{2}\Gamma_1)t\right) \pm \exp\left(-i(m_2 - \frac{i}{2}\Gamma_2)t\right)\right]. \tag{4}$$

The average mass and width of the two mass eigenstates, and their differences are defined by:

$$\begin{aligned} m_1 + m_2 = 2m,\ m_1 - m_2 = \Delta m, \\ \Gamma_1 + \Gamma_2 = 2\Gamma,\ \Gamma_1 - \Gamma_2 = \Delta\Gamma. \end{aligned} \tag{5}$$

The $g_\pm(t)$ coefficients can then be written as:

$$\begin{aligned} g_+(t) &= \exp\left(-(im + \frac{\Gamma}{2})t\right)\cosh\left[(i\Delta m + \frac{\Delta\Gamma}{2})\frac{t}{2}\right], \\ g_-(t) &= \exp\left(-(im + \frac{\Gamma}{2})t\right)\sinh\left[-(i\Delta m + \frac{\Delta\Gamma}{2})\frac{t}{2}\right]. \end{aligned} \tag{6}$$

If Δm and $\Delta\Gamma$ vanish, there will be no mixing effects as shown in Eq. 2. Usually the mixing paramters are defined by:

$$x \equiv \frac{\Delta m}{\Gamma},\ y \equiv \frac{\Delta\Gamma}{2\Gamma}. \tag{7}$$

With Eq. 2, the transition amplitudes of initial pure D^0 and $\bar{D}^0$ mesons decaying to final state f are then obtained to be:

$$\begin{aligned} \langle f|D^0(t)\rangle &= g_+(t)A_f + \tfrac{q}{p} g_-(t)\bar{A}_f, \\ \langle f|\bar{D}^0(t)\rangle &= g_+(t)A_f + \tfrac{p}{q} g_-(t)\bar{A}_f. \end{aligned} \tag{8}$$

by denoting $\langle f|D^0\rangle = A_f$ and $\langle f|\bar{D}^0\rangle = \bar{A}_f$. It is convenient to define:

$$\rho_f = \frac{\bar{A}_f}{A_f} = r_D^f \exp\left(-i\delta_D^f\right) \text{ and } \lambda_f = \frac{q}{p}\rho_f = r_{CP} r_D^f \exp\left(-i(\delta_D^f - \phi)\right), \tag{9}$$

where δ_D^f is the *CP*-conserving phase (strong-phase difference) in the decay and ϕ is the *CP*-violating phase that arises from charm mixing. The analogous formula for the *CP*-conjugate decay is:

$$\bar{\rho}_{\bar{f}} = \frac{A_{\bar{f}}}{\bar{A}_{\bar{f}}} = r_D^f \exp\left(-i\delta_D^f\right) \text{ and } \lambda_f = \frac{p}{q}\rho_f = r_{CP}^{-1} r_D^f \exp\left(-i(\phi + \delta_D^f)\right), \tag{10}$$

the time-dependent decay rate can then be expressed as:

$$\begin{aligned}
\Gamma(D^0(t) \to f) \propto |A_f|^2 \exp\left(-\Gamma t\right) \; [\; & C_y \cosh(y\Gamma t) + C_x \cos(x\Gamma t) \\
& + S_y \sinh(y\Gamma t) - S_x \sin(x\Gamma t)], \\
\Gamma(\bar{D}^0(t) \to f) \propto |\bar{A}_f|^2 \exp\left(-\Gamma t\right) \; [\; & \bar{C}_y \cosh(y\Gamma t) + \bar{C}_x \cos(x\Gamma t) \\
& + \bar{S}_y \sinh(y\Gamma t) - \bar{S}_x \sin(x\Gamma t)].
\end{aligned} \tag{11}$$

Here,

$$\begin{aligned}
C_y &\equiv \frac{1 + |\lambda_f|^2}{2}, \; S_y \equiv -\Re(\lambda_f), \\
C_x &\equiv \frac{1 - |\lambda_f|^2}{2}, \; S_x \equiv -\Im(\lambda_f),
\end{aligned} \tag{12}$$

and

$$(\bar{C}_y, \; \bar{S}_y, \; \bar{C}_x, \; \bar{S}_x) = \left|\frac{p}{q}\right|^2 (C_y, \; S_y, \; -C_x, \; -S_x). \tag{13}$$

The time-integrated decay rate is then given by:

$$\begin{aligned}
\Gamma(D^0(t \to \infty) \to f) &\propto |A_f|^2 \left[\tfrac{1}{1-y^2} C_y + \tfrac{1}{1+x^2} C_x + \tfrac{y}{1-y^2} S_y - \tfrac{x}{1+x^2} S_x\right], \\
\Gamma(\bar{D}^0(t \to \infty) \to f) &\propto |A_f|^2 \left[\tfrac{1}{1-y^2} \bar{C}_y + \tfrac{1}{1+x^2} \bar{C}_x + \tfrac{y}{1-y^2} S_y - \tfrac{x}{1+x^2} \bar{S}_x\right].
\end{aligned} \tag{14}$$

Experimental sensitivity on the mixing parameters has shown that it's safe to ignore the higher order terms up to $o(x^2, \; y^2)$. With this approximation, Eq. 11 becomes

$$\begin{aligned}
\Gamma(D^0(t) \to f) \propto |A_f|^2 \exp\left(-\Gamma t\right) \; [\; & 1 + \frac{1}{4}(x^2 + y^2)|\lambda_f|^2 \Gamma^2 t^2 \\
& - \frac{1}{4}(x^2 - y^2)\Gamma^2 t^2 - (y\Re(\lambda_f) - x\Im(\lambda_f))\Gamma t)], \\
\Gamma(\bar{D}^0(t) \to f) \propto |\bar{A}_f|^2 \left|\frac{p}{q}\right|^2 \exp\left(-\Gamma t\right) \; [\; & |\lambda_f|^2 + \frac{1}{4}(x^2 + y^2)\Gamma^2 t^2 \\
& - \frac{1}{4}(x^2 - y^2)|\lambda_f|^2 \Gamma^2 t^2 - (y\Re(\lambda_f) + x\Im(\lambda_f))\Gamma t)].
\end{aligned} \tag{15}$$

The corresponding decay rates for $D^0(t)/\bar{D}^0(t) \to \bar{f}$ decays are:

$$\begin{aligned}
\Gamma(D^0(t) \to \bar{f}) \propto |\bar{A}_{\bar{f}}|^2 \left|\frac{q}{p}\right|^2 \exp\left(-\Gamma t\right) \; [\; & |\bar{\lambda}_{\bar{f}}|^2 + \frac{1}{4}(x^2 + y^2)\Gamma^2 t^2 \\
& - \frac{1}{4}(x^2 - y^2)|\bar{\lambda}_{\bar{f}}|^2 \Gamma^2 t^2 - (y\Re(\bar{\lambda}_{\bar{f}}) + x\Im(\bar{\lambda}_{\bar{f}}))\Gamma t)], \\
\Gamma(\bar{D}^0(t) \to \bar{f}) \propto |\bar{A}_{\bar{f}}|^2 \exp\left(-\Gamma t\right) \; [\; & |1 + \frac{1}{4}(x^2 + y^2)|\bar{\lambda}_{\bar{f}}|^2 \Gamma^2 t^2 \\
& - \frac{1}{4}(x^2 - y^2)\Gamma^2 t^2 - (y\Re(\bar{\lambda}_{\bar{f}}) - x\Im(\bar{\lambda}_{\bar{f}}))\Gamma t)].
\end{aligned} \tag{16}$$

Eq. 15 and 16, ie. the time-integrated decay rates, becomes

$$\begin{aligned}
\Gamma(D^0 \to f) &\propto |A_f|^2 \left[1 + \frac{1}{2}(x^2+y^2)|\lambda_f|^2 - \frac{1}{2}(x^2-y^2) - (y\Re(\lambda_f) - x\Im(\lambda_f))\right], \\
\Gamma(\bar{D}^0 \to f) &\propto |A_f|^2 \left[|\lambda_f|^2 + \frac{1}{2}(x^2+y^2) - \frac{1}{2}(x^2-y^2)|\lambda_f|^2 - (y\Re(\lambda_f) + x\Im(\lambda_f))\right], \\
\Gamma(D^0 \to \bar{f}) &\propto |\bar{A}_{\bar{f}}|^2 \left[|\bar{\lambda}_{\bar{f}}|^2 + \frac{1}{2}(x^2+y^2) - \frac{1}{2}(x^2-y^2)|\bar{\lambda}_{\bar{f}}|^2 - (y\Re(\bar{\lambda}_{\bar{f}}) + x\Im(\bar{\lambda}_{\bar{f}}))\right], \\
\Gamma(\bar{D}^0 \to \bar{f}) &\propto |\bar{A}_{\bar{f}}|^2 \left[1 + \frac{1}{2}(x^2+y^2)|\bar{\lambda}_{\bar{f}}|^2 - \frac{1}{2}(x^2-y^2) - (y\Re(\bar{\lambda}_{\bar{f}}) + x\Im(\bar{\lambda}_{\bar{f}}))\right].
\end{aligned} \tag{17}$$

At CLEO and BESIII experiments, large data samples of $D\bar{D}$ pairs have been accumulated, corresponding to integrated luminosities of 818 nb^{-1} and 20 fb^{-1} at the resonance of $\psi(3770)$ respectively. The pair production enables absolute measurement of the decay branching fractions (BF) and study of the decay amplitudes. In the past decade, progresses have also been made in determination of the strong-phase differences in D^0 and $\bar{D}^0$ decays to the same final state. Such studies have been performed base on a double tag (DT) method, where both of the two charm mesons are reconstructed. For the neutral D meson decays studied with the DT method, the quantum correlation (QC) between $D\bar{D}$ mesons should be considered.

The QC effects arises from the fact that, at the quark level, $e^+e^- \to c\bar{c}$ proceeds through a virtual photon with $J^{PC} = 1^{--}$. Therefore the initially produced $D^0\bar{D}^0$ mesons at the $\psi(3770)$ resonance are in a QC state with a finite C-odd parity. At higher energy points, accompanying particles of π^0s and photons can be emitted. The C parity is even (odd) when the number of photons is odd (even). There could be initial state radiation or two-photon processes that dilutes the C correlation. However their contributions are negligible(?). As a result, when one neutral D meson decays to a known CP eigenstate, the other D decay is also a CP eigenstate.

By exploiting the quantum correlations, the CP violation parameter $\frac{q}{p}$ and mixing parameters can be determined from the time-integrated double-tag decay rates. It is noted that near threshold the boost of charm mesons is not large enough to perform time-dependent measurements. However the quantum correlation effects would be of particular importance to such studies as it complement to the time-dependent measurement. Therefore it is useful to study the $D\bar{D}$ double decaying to the final states, eg. $(f_1,\ f_2)$, $(f_1,\ \bar{f}_2)$, $(\bar{f}_1,\ f_2)$ and $(\bar{f}_1,\ \bar{f}_2)$.

The correlated $D\bar{D}$ state can be written as

$$\frac{1}{\sqrt{2}}\left[|D^0(\mathbf{K},\ t)\rangle\,|\bar{D}^0(-\mathbf{K},\ t)\rangle + C\,|D^0(-\mathbf{K},\ t)\rangle\,|\bar{D}^0(\mathbf{K},\ t)\rangle\right], \tag{18}$$

where $C = \pm 1$ denotes the charge-conjugation parity. When the D meson with momentum $\mathbf{K}$ decays to f_1 at time t_1 and the other with momentum $-\mathbf{K}$ decays to f_2 at time t_2, the decay rate is given by

$$\begin{aligned}
A(f_1,t_1;f_2,t_2)_C = &\frac{1}{\sqrt{2}}A_{f_1}A_{f_2}\xi_C\left[g_+(t_1)g_-(t_2) + Cg_-(t_1)g_+(t_2)\right] \\
&+ \frac{1}{\sqrt{2}}A_{f_1}A_{f_2}\zeta_C\left[g_+(t_1)g_+(t_2) + Cg_-(t_1)g_-(t_2)\right],
\end{aligned} \tag{19}$$

where

$$\begin{aligned}
\xi_C &= \frac{p}{q}(1 + C\lambda_{f_1}\lambda_{f_2}), \\
\zeta_C &= \frac{p}{q}(\lambda_{f_2} + C\lambda_{f_1}).
\end{aligned} \tag{20}$$

The time-dependent decay rate can then be written as

$$\begin{aligned}
A(f_1,t_1;f_2,t_2)_C \propto\ & A_{f_1}^2 A_{f_2}^2 \exp(-\Gamma t_C) \\
& \left[(|\xi_C|^2 + |\zeta_C|^2)\cosh(y\Gamma t_C) - 2\Re(\xi_C^*\zeta_C)\sinh(y\Gamma t_C)\right. \\
& \left. -(|\xi_C|^2 - |\zeta_C|^2)\cos(x\Gamma t_C) - 2\Im(\xi_C^*\zeta_C)\sin(x\Gamma t_C)\right],
\end{aligned} \tag{21}$$

where $t_C = t_2 + Ct_1$. The time-integrated decay rate can be obtained from Eq. 21 to be:

$$\Gamma(f_1; f_2)_C \propto A_{f_1}^2 A_{f_2}^2 \left[\frac{1+Cy^2}{(1-y^2)^2}(|\xi_C|^2 + |\zeta_C|^2) - \frac{2(1+C)y}{(1-y^2)^2}\Re(\xi_C^*\zeta_C) \right.$$
$$\left. -\frac{1-Cx^2}{(1+x^2)^2}(|\xi_C|^2 - |\zeta_C|^2) - \frac{2(1+C)x}{(1+x^2)^2\Im(\xi_C^*\zeta_C)} \right] \tag{22}$$

It it noted that for C-odd correlated $D\bar{D}$ decays, the terms linear to the mixing parameters vanish, indicating that experimentally C-even correlated $D\bar{D}$ dataset is superior to charm mixing and CP violation studies, whereas C-odd correlated $D\bar{D}$ pairs is also essential to determination of the strong-phase parameters in charm meson decays without significant effects from charm mixing.

Again, by ignoring the higher order terms up to $o(x^2,\ y^2)$, and according to Eq. 22, the double decay rates can be written by:

$$\begin{aligned}
\Gamma(f_1; f_2)_C \propto\ & A_{f_1}^2 A_{f_2}^2 \left|\frac{p}{q}\right|^2 \left\{(2+C)(x^2+y^2)|1+C\lambda_{f_1}\lambda_{f_2}|^2 \right. \\
& + \left[2-(2+C)(x^2-y^2)\right]|\lambda_{f_2}+C\lambda_{f_1}|^2 \\
& -2(1+C)y\left[(1+|\lambda_{f_1}|^2)\Re(\lambda_{f_2}) + C(1+|\lambda_{f_2}|^2)\Re(\lambda_{f_1})\right] \\
& \left. -2(1+C)x\left[(1-|\lambda_{f_1}|^2)\Im(\lambda_{f_2}) + C(1-|\lambda_{f_2}|^2)\Im(\lambda_{f_1})\right]\right\} \\
\Gamma(\bar{f}_1; \bar{f}_2)_C \propto\ & A_{\bar{f}_1}^2 A_{\bar{f}_2}^2 \left|\frac{q}{p}\right|^2 \left\{(2+C)(x^2+y^2)|1+C\bar{\lambda}_{\bar{f}_1}\bar{\lambda}_{\bar{f}_2}|^2 \right. \\
& + \left[2-(2+C)(x^2-y^2)\right]|\bar{\lambda}_{\bar{f}_2}+C\bar{\lambda}_{\bar{f}_1}|^2 \\
& -2(1+C)y\left[(1+|\bar{\lambda}_{\bar{f}_1}|^2)\Re(\bar{\lambda}_{\bar{f}_2}) + C(1+|\bar{\lambda}_{\bar{f}_2}|^2)\Re(\bar{\lambda}_{\bar{f}_1})\right] \\
& \left. -2(1+C)x\left[(1-|\bar{\lambda}_{f_1}|^2)\Im(\bar{\lambda}_{f_2}) + C(1-|\bar{\lambda}_{f_2}|^2)\Im(\bar{\lambda}_{f_1})\right]\right\}
\end{aligned} \tag{23}$$

and

$$\begin{aligned}
\Gamma(f_1; \bar{f}_2)_C \propto\ & A_{f_1}^2 \bar{A}_{\bar{f}_2}^2 \left\{(2+C)(x^2+y^2)|\bar{\lambda}_{\bar{f}_2}+C\lambda_{f_1}|^2 \right. \\
& + \left[2-(2+C)(x^2-y^2)\right]|1+C\lambda_{f_1}\bar{\lambda}_{\bar{f}_2}|^2 \\
& -2(1+C)y\left[(1+|\lambda_{f_1}|^2)\Re(\bar{\lambda}_{\bar{f}_2}) + C(1+|\bar{\lambda}_{\bar{f}_2}|^2)\Re(\lambda_{f_1})\right] \\
& \left. +2(1+C)x\left[(1-|\lambda_{f_1}|^2)\Im(\bar{\lambda}_{\bar{f}_2}) + C(1-|\bar{\lambda}_{\bar{f}_2}|^2)\Im(\lambda_{f_1})\right]\right\} \\
\Gamma(\bar{f}_1; f_2)_C \propto\ & \bar{A}_{\bar{f}_1}^2 A_{f_2}^2 \left\{(2+C)(x^2+y^2)|\lambda_{f_2}+C\bar{\lambda}_{\bar{f}_1}|^2 \right. \\
& + \left[2-(2+C)(x^2-y^2)\right]|1+C\bar{\lambda}_{\bar{f}_1}\lambda_{f_2}|^2 \\
& -2(1+C)y\left[(1+|\bar{\lambda}_{\bar{f}_1}|^2)\Re(\lambda_{f_2}) + C(1+|\lambda_{f_2}|^2)\Re(\bar{\lambda}_{\bar{f}_1})\right] \\
& \left. +2(1+C)x\left[(1-|\bar{\lambda}_{\bar{f}_1}|^2)\Im(\lambda_{f_2}) + C(1-|\lambda_{f_2}|^2)\Im(\bar{\lambda}_{\bar{f}_1})\right]\right\}
\end{aligned} \tag{24}$$

In LHCb and Belle (II) experiments, mixing and indirect CPV in the neutral D meson system are studied in a time-dependent way according to Eq.16. It requires a large boost of momentum for the D meson. The LHCb experiment has made the most precise measurement of the width difference parameter y and mass difference parameter x up to date in decays K^+K^- and $\pi^+\pi^-$ decays [1], and $K_S^0\pi^+\pi^-$ decay [2] respectively, which also give the measurement of the CP violation in the charm sector. The direct CP violation in charm decays has been observed for the first time by investigating the difference of the decay asymmetries in K^+K^- and $\pi^+\pi^-$ decays [3]. However the indirect CP violation associated with charm mixing hasn't been observed.

There is no meaningful time-integrated measurement of x and y at CLEO and BESIII experiments [4, 5]. It would be desired to perform such measurements in different decay channels and laboratories as cross checks with the time-dependent analyses. The planned STCF, where large data samples of C-even $D\bar{D}$ would be accumulated, provides a good opportunity to determination of x, y, $|q/p|$ and ϕ. The direct CP violation can also be studied. In this report, we first demonstrate the generation of simulated QC $D\bar{D}$ decays followed by the formalism and methodology of the experimental studies with a series of decays. We also present the sensitivity of the charm mixing and CP violation studies in STCF based on the simulated events.

II. SIMULATION OF QC $D\bar{D}$ DECAYS

The *EvtGen* package is used in BESIII and other experiments to generate MC samples containing the $D\bar{D}$ decays. However it is hard to simulate the QC effects directly within the current software framework for inclusive D decays because the *CP*-conserving phases, denoted as strong phases, vary in these different decays. The QC double decay rate where D decays to f_1 and $\bar{D}$ decays to f_2 in Eqs. 23 and 24 can be rewritten as a function of the strong-phase differences δ_D^f

$$\begin{aligned} R(f_1;\bar{f}_2) \propto |A_{f_1}|^2|A_{f_2}|^2\{&(2+C)(x^2+y^2)[(r_{CP})^2(r_D^{f_1})^2+(r_{CP}^{-1})^2(r_D^{f_2})^2+2Cr_D^{f_1}r_D^{f_2}\cos(\delta_D^{f_1}-\delta_D^{f_2}-2\phi)] \\ &+[2-(2+C)(x^2-y^2)][1+(r_D^{f_1})^2(r_D^{f_2})^2+2Cr_D^{f_1}r_D^{f_2}\cos(\delta_D^{f_1}+\delta_D^{f_2})] \\ &-2(1+C)y[r_D^{f_2}(r_{CP}^{-1}+r_{CP}(r_D^{f_1})^2)\cos(\delta_D^{f_2}+\phi)+Cr_D^{f_1}(r_{CP}+r_{CP}^{-1}(r_D^{f_2})^2)\cos(\delta_D^{f_1}-\phi)] \\ &-2(1+C)x[r_D^{f_2}(r_{CP}^{-1}-r_{CP}(r_D^{f_1})^2)\sin(\delta_D^{f_2}+\phi)+Cr_D^{f_1}(r_{CP}-r_{CP}^{-1}(r_D^{f_2})^2)\sin(\delta_D^{f_1}-\phi)]\}. \end{aligned} \tag{25}$$

For a simulated double D decay event, the initial uncorrelated amplitude $|A_{f_1}|^2|A_{f_2}|^2$ should be corrected with the factor of QC as shown in the above function. The D decay parameters can be derived from the amplitude model that is used to simulate the MC event. The charm mixing ad indirect *CP* violation parameters are difficult to calculate in theory so that they should be taken from existing experimental measurements to obtain the correction factor of QC. The mode-specific direct *CP* violation has been neglected therefore it can be tested under null hypotheses with the corrected MC samples and the sensitivity of direct *CP* violation can be obtained.

A tool of *QCMCFilterAlg* has been developed to correct the MC samples of inclusive $D\bar{D}$ decays event-by-event. When the final state of D decay is a *CP*-even(odd) eigenstate, the strong-phase difference $\delta_D^f = 0(\pi)$ and $r_D^f = 1$. Two-body systems of particle and anti-particle like K^+K^-, $\pi^+\pi^-$ are in a *CP*-even state as they are invariant under *CP* transformation. If the final-state particles are all neutral spinless *CP* eigenstates, the *CP* parity can be determined by the product of their *CP* eigenvalues. We have neglected the *CP* violation in neutral kaon system, namingly K_S^0 and K_L^0 have been approximated to be *CP* even and odd respectively. r_D^f of the flavour-specific (semi)leptonic decays equals to zero so that the related terms all vanish. Other decays are not *CP* eigenstates that the QC correction factor can be determined with the model predictions of the strong-phase differences and amplitude ratios. The decays which are non-*CP* eigenstates and have been included in the package consists of quasi-flavour decays of $K^-\pi^+$, $K^-\pi^+\pi^0$ and $K^-\pi^+\pi^+\pi^-$, self-conjugate decays of $K_{S,L}^0\pi^+\pi^-$, $K_{S,L}^0K^+K^-$, $\pi^+\pi^-\pi^+\pi^-$, $\pi^+\pi^-\pi^0\pi^0$, $K^+K^-\pi^+\pi^-$, $\pi^+\pi^-\pi^0$ and $K_{S,L}^0\pi^+\pi^-\pi^0$. The strong-phase differences and amplitude ratios have been calculated with the same amplitude models as those used to simulate the decay events. The above modes are leading as to the studies of charm mixing and *CP* violation and determination of the *CP*-violating CKM angle γ. It is found to be sufficient to have them in the package for the toy studies presented in the report and they are used to estimate the background contributions to the model-independent measurement of the strong-phase parameters.

Inclusive samples have been generated at the center-of-mass energy point $\sqrt{s} = 4040$ MeV, where the sensitivity of charm mixing and *CP* violation studies are found to be optimal. $D^{(*)0}\bar{D}^{*0}$ and $D^{(*)-}\bar{D}^{*+}$ pairs are simulated. As D^* can decay to π^0D and γD, the neutral $D\bar{D}$ is in a C-even(odd) correlated state when the number of transition photons is odd(even). Flavour-specific D decays can be obtained with the $D^{(*)-}\bar{D}^{*+}$ pairs by reconstructing the soft accompanying charged pion. With the parameters, the correction factor is determined for each double decay event and is normalized by the maximal factor, which is ? if both D mesons decay into the same (opposite) *CP* eigenstates for C-even (odd) correlated $D\bar{D}$ pair. A uniformly distributed random number in the range 0-1 is then generated. The event is kept (discarded) when the correction factor is smaller (larger) than the random number.

III. FORMALISM

- In the case of c-odd qc
- In the case of c-even qc

IV. PERFORMANCE

- MC un-qc vs MC qc vs Data

V. CONCLUSION/SUMMARY

- The qcfilter package is improved to model c-even, c-odd qc and include newly measured models
- not strongly depend on the software framework, can be implemented to other experiments or theorists
- prepared for new measurements (20/fb 3770, c-even on stcf)

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